

Alaoglu–Erdős After the Unified Report

Rational-Orbit Denominators, Farey Entropy, Smooth Rigidity, and an Audited Attempt at the Remaining Proof

Research continuation prepared for Vladimir Reshetnikov

August 21, 2026

Abstract

The Alaoglu–Erdős conjecture asserts that a real number x satisfying $2^x, 3^x \in \mathbb{Z}$ must be an integer. The supplied unified report is already a very large research dossier: it develops exact conditional structure, determinant methods, matrix and operator reformulations, p -adic diagnostics, approximation barriers, and extensive formal verification. The present document is a separate continuation rather than a restatement of that dossier.

The new work concentrates on a route suggested by the July 2026 theorem of Lelis, Moreira, and Silva on smooth functions mapping rationals to rationals with controlled denominators. Assuming a counterexample, we first sharpen the compact-orbit description to an exact numerator, denominator, and height law. We then prove a new *Farey-to-orbit entropy theorem*: every injective rational parametrization of the dense conditional orbit must distort denominators at least like $\exp(cQ^2)$ on rationals of height at most Q . Consequently, no polynomially tame reparametrization can place the orbit in the regime of known rational-mapping rigidity theorems. We also derive an orbit-adapted subunit-locking inequality and show quantitatively why every fixed-order rational-contact or divided-difference certificate is underpowered: its analytic gain is only polynomial in the orbit index, whereas its arithmetic denominator cost is exponential.

These results rigorously close several plausible proof routes and identify a precise missing amplification principle. They do not supply an unconditional proof of the conjecture. Current transcendence theory still places the decisive 2×2 logarithmic determinant at the Four Exponentials boundary; the recent Matrix Coefficient Conjecture is equivalent to Four Exponentials in size two and remains conjectural. No statement in this report is promoted beyond the proof actually given.

Proof-status statement. The report proves all numbered elementary and height-theoretic results below, including the exact orbit-height formula and the Farey-to-orbit entropy theorem. It does *not* relabel an unresolved implication as a proof of the conjecture. The final size-two logarithmic step remains open in the verified literature; every conditional input is marked explicitly.

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1 Scope, non-duplication, and the logical standard

1.1 What was taken from the supplied report

The source file

`/Article/alaoglu-erdos-unified-report/alaoglu-erdos-unified-report.tex`

contains roughly nineteen thousand lines and already consolidates a large collection of independent continuations. In particular, it contains:

- the rational-exponent reduction and the Gelfond–Schneider consequence;
- the exact logarithmic-determinant and Four Exponentials formulation;
- an exact conditional monoid of solutions, a primitive generator, and compact synchronized dyadic–triadic orbits;
- determinant, generalized Vandermonde, HCIZ, Smith-divisor, finite-difference, approximation, matrix, spectral, Mahler, and operator analyses;
- exact reduced denominator exponents, qualitative and asymptotic compact-height counting, and rational-contact transversality;
- an extensive catalogue of failure points and a literature audit through most of 2026.

Those arguments are not reproduced. We use only a minimal baseline and independently prove the refinements needed here. The genuinely new core of this continuation is:

1. an exact closed formula for the heights of every point in the compact conditional orbit, not merely an $\asymp$ estimate;
2. a Farey-counting theorem showing that any injection from all rationals in an interval into that orbit has denominator distortion $\exp(cQ^2)$;
3. a corresponding no-go theorem for polynomially tame rational reparametrizations and for direct use of the 2026 smooth rational-map rigidity theorem;
4. an orbit-adapted comparison between arithmetic lower bounds and analytic contact estimates, isolating the exact exponential-versus-polynomial scale mismatch;
5. a set of proof-producing targets stated strongly enough that crossing any one of them would be a genuine advance rather than another reformulation.

1.2 Current status of the transcendence input

Dasgupta and Kakde formulate the Matrix Coefficient Conjecture for matrices of logarithms of algebraic numbers and explicitly note that in dimension two it is equivalent to the Four Exponentials Conjecture [4]. Their paper proves several implications and builds a support-sensitive strategy, but leaves a key implication open. Thus it cannot be cited as a proof of the determinant nonvanishing needed here.

The recent paper of Lelis, Moreira, and Silva proves a powerful rigidity theorem for C^{2k+1} functions mapping every rational in a neighborhood to a rational with denominator $O(q^k)$, provided a rational jet through order $2k$ is available [5]. This theorem is highly relevant to the compact rational orbit, but two hypotheses fail in a sharp way: the conditional graph contains only logarithmically many points of height at most H , and its first derivative at every rational graph point is transcendental. Sections 4–5 quantify the first failure and make the second explicit.

1.3 What counts as a proof

A proof must derive a contradiction from only

$$x \in \mathbb{R}, \quad 2^x \in \mathbb{Z}, \quad 3^x \in \mathbb{Z},$$

using established theorems. The following are not proofs:

- assuming Four Exponentials, Schanuel, or the Matrix Coefficient Conjecture without labeling the assumption;
- replacing exact rationality by numerical near-rationality;
- invoking density of a rational orbit as though a continuous function mapping a dense rational set to rationals had to be rational;
- using an arithmetic lower bound e^{-cN} against an analytic upper bound N^{-d} and reversing their asymptotic order;
- treating real logarithmic proportionality as a p -adic proportionality without first proving continuity of the induced homomorphism.

The scale comparison in this report was designed specifically to expose these errors before they enter a purported proof.

2 Minimal reduction to the exact logarithmic wall

Proposition 2.1 (Elementary reduction). *Suppose $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$. Then $x \geq 0$. If $x \in \mathbb{Q}$, then $x \in \mathbb{N}_0$. If $x \notin \mathbb{Z}$, then x is transcendental.*

Proof. If $x < 0$, then $0 < 2^x < 1$, so 2^x is not an integer. Now write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{Z}$, then $2^a = m^b$. Taking the 2-adic valuation gives $a = bv_2(m)$, so $b \mid a$. Coprimality forces $b = 1$.

If x were algebraic and irrational, Gelfond–Schneider would make 2^x transcendental, contrary to $2^x \in \mathbb{Z}$. Hence every nonintegral candidate is transcendental. $\square$

Put

$$m = 2^x, \quad n = 3^x.$$

Then

$$\log m \log 3 - \log n \log 2 = 0. \tag{1}$$

Equivalently,

$$\det \begin{pmatrix} \log 2 & \log 3 \\ \log m & \log n \end{pmatrix} = 0.$$

Since $\log m = x \log 2$ and $\log n = x \log 3$, this determinant vanishes tautologically under the hypotheses. The issue is whether a singular matrix of logarithms of algebraic numbers can have both row and column directions rationally independent.

Proposition 2.2 (Conditional Four Exponentials proof). *The Four Exponentials Conjecture implies the Alaoglu–Erdős conjecture.*

Proof. Assume $x \notin \mathbb{Z}$. Proposition 2.1 gives $x \notin \mathbb{Q}$, so $1, x$ are linearly independent over $\mathbb{Q}$. The numbers $\log 2, \log 3$ are also linearly independent over $\mathbb{Q}$, since $2^a 3^b = 1$ with $a, b \in \mathbb{Z}$ implies $a = b = 0$. The four exponentials

$$e^{\log 2} = 2, \quad e^{\log 3} = 3, \quad e^{x \log 2} = m, \quad e^{x \log 3} = n$$

are all algebraic. Four Exponentials predicts that at least one must be transcendental, a contradiction. $\square$

This proposition is not an unconditional proof because Four Exponentials is open. The rest of the report asks whether the stronger fact that all four exponentials are positive integers supplies an independent route around that conjecture.

3 The compact conditional orbit with exact heights

Assume for the remainder of Sections 3–6 that a nonintegral counterexample exists. Write

$$x = B + \alpha, \quad B = \lfloor x \rfloor \in \mathbb{N}_0, \quad 0 < \alpha < 1.$$

By Proposition 2.1, α is irrational. Set

$$U = 2^\alpha = \frac{2^x}{2^B} \in \mathbb{Q} \cap (1, 2), \quad V = 3^\alpha = \frac{3^x}{3^B} \in \mathbb{Q} \cap (1, 3).$$

Write these rationals in lowest terms as

$$U = \frac{W}{2^r}, \quad W > 1 \text{ odd}, \quad r \geq 1, \quad (2)$$

$$V = \frac{Z}{3^s}, \quad Z > 1, \quad 3 \nmid Z, \quad s \geq 0. \quad (3)$$

The denominator restrictions follow because U is an integer divided by a power of 2, and similarly for V . Moreover,

$$r + \alpha = \log_2 W, \quad s + \alpha = \log_3 Z. \quad (4)$$

For $k \geq 0$, define

$$R_k = 2^{\{k\alpha\}}, \quad S_k = 3^{\{k\alpha\}}. \quad (5)$$

The set $\{\{k\alpha\} : k \geq 0\}$ is dense in $[0, 1)$, so (R_k, S_k) is dense on the analytic arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

The next theorem gives exact arithmetic data for every orbit point.

Theorem 3.1 (Exact reduced orbit law). *For every integer $k \geq 1$, the following are lowest-term expressions:*

$$R_k = \frac{W^k}{2^{\lfloor k \log_2 W \rfloor}}, \quad (6)$$

$$S_k = \frac{Z^k}{3^{\lfloor k \log_3 Z \rfloor}}. \quad (7)$$

Consequently,

$$H(R_k) = W^k, \quad \text{den}(R_k) = 2^{\lfloor k \log_2 W \rfloor}, \quad (8)$$

$$H(S_k) = Z^k, \quad \text{den}(S_k) = 3^{\lfloor k \log_3 Z \rfloor}. \quad (9)$$

In particular,

$$\frac{W^k}{2} < \text{den}(R_k) < W^k, \quad \frac{Z^k}{3} < \text{den}(S_k) < Z^k. \quad (10)$$

Proof. From $U = W/2^r = 2^\alpha$,

$$R_k = 2^{k\alpha - \lfloor k\alpha \rfloor} = \frac{W^k}{2^{rk + \lfloor k\alpha \rfloor}}.$$

Equation (4) gives

$$rk + \lfloor k\alpha \rfloor = \lfloor k(r + \alpha) \rfloor = \lfloor k \log_2 W \rfloor.$$

The numerator is odd, so the fraction is reduced. The proof of (7) is identical, using $3 \nmid Z$.

Because $1 < R_k < 2$, the numerator of the reduced fraction is larger than its denominator and smaller than twice its denominator. Hence its projective height is exactly W^k , and $W^k/2 < \text{den}(R_k) < W^k$. The base-3 statement follows from $1 < S_k < 3$. $\square$

Corollary 3.2 (Exact compact-orbit counting). *Let*

$$H_k = \max\{H(R_k), H(S_k)\}, \quad C_0 = \max\{W, Z\}.$$

Then $H_k = C_0^k$. *Hence, for* $H \geq 1$,

$$\#\{k \geq 0 : H_k \leq H\} = 1 + \left\lfloor \frac{\log H}{\log C_0} \right\rfloor. \quad (11)$$

Proof. The height identity follows from Theorem 3.1. The points are distinct because α is irrational. The inequality $C_0^k \leq H$ is equivalent to $k \leq \log H / \log C_0$. $\square$

The supplied report proves the qualitative dichotomy 1 versus $\asymp \log H$ for the compact arc. Corollary 3.2 records the precise conditional counting function and is the input needed for the entropy argument below.

4 The Farey-to-orbit entropy mismatch

The natural hope inspired by rational-map rigidity is to reparametrize the dense orbit by all rationals in an interval, then apply a theorem about smooth functions preserving rationality. This section proves that every such reparametrization is arithmetically violent.

Fix a nondegenerate closed interval $I = [a, b] \subset (0, 1)$ and write $|I| = b - a$. Let

$$\mathcal{F}_I(Q) = \left\{ \frac{p}{q} \in I : 1 \leq q \leq Q, \gcd(p, q) = 1 \right\}.$$

The classical Farey estimate is

$$\#\mathcal{F}_I(Q) = \frac{3|I|}{\pi^2} Q^2 + O_I(Q \log Q). \quad (12)$$

A proof sketch is included in Appendix A.

Definition 4.1 (Orbit index). For a point $P = (R_k, S_k)$ in the one-sided compact orbit, write $\nu(P) = k$. This is well-defined because the orbit points are distinct.

Theorem 4.2 (Farey-to-orbit entropy theorem). *Let*

$$\Phi : \mathbb{Q} \cap I \longrightarrow \{(R_k, S_k) : k \geq 0\}$$

be injective. Define

$$D_2(Q) = \max_{t \in \mathcal{F}_I(Q)} \text{den}(\pi_1 \Phi(t)), \quad D_3(Q) = \max_{t \in \mathcal{F}_I(Q)} \text{den}(\pi_2 \Phi(t)).$$

Then

$$\liminf_{Q \rightarrow \infty} \frac{\log D_2(Q)}{Q^2} \geq \frac{3|I|}{\pi^2} \log W, \quad (13)$$

$$\liminf_{Q \rightarrow \infty} \frac{\log D_3(Q)}{Q^2} \geq \frac{3|I|}{\pi^2} \log Z. \quad (14)$$

In particular, for every fixed $A > 0$, neither $D_2(Q)$ nor $D_3(Q)$ is $O(Q^A)$.

Proof. Put $N_Q = \#\mathcal{F}_I(Q)$. The images of these N_Q rationals have distinct orbit indices. Therefore at least one image has index at least $N_Q - 1$; equivalently,

$$\max_{t \in \mathcal{F}_I(Q)} \nu(\Phi(t)) \geq N_Q - 1.$$

By Theorem 3.1, an orbit point of index k has first-coordinate denominator greater than $W^k/2$ and second-coordinate denominator greater than $Z^k/3$. Consequently,

$$D_2(Q) > \frac{1}{2} W^{N_Q-1}, \quad D_3(Q) > \frac{1}{3} Z^{N_Q-1}.$$

Taking logarithms, dividing by Q^2 , and using (12) proves the two limits. Since $\log Q^A/Q^2 \rightarrow 0$, no polynomial bound is possible. $\square$

Remark 4.3. The theorem uses only injectivity and the exact orbit-height law. It does not use continuity, differentiability, order preservation, or equidistribution. Thus adding analytic regularity to the parametrization cannot remove the arithmetic distortion.

Corollary 4.4 (No polynomially tame rational conjugacy). *Let $J = (a, b)$ be a nonempty open interval, and suppose an increasing homeomorphism $\phi : J \rightarrow (1, 2)$ satisfies*

$$\phi(\mathbb{Q} \cap J) = \{R_k : k \geq 1\}.$$

For each rational $t \in J$, let $G(t) = S_k$ where $\phi(t) = R_k$. Then G is the restriction to $\mathbb{Q} \cap J$ of the continuous function

$$t \mapsto \phi(t)^{\log_2 3}.$$

Moreover, for every nondegenerate closed interval $I \Subset J$,

$$\max_{p/q \in \mathcal{F}_I(Q)} \text{den } G(p/q) \geq \exp\left(\left(\frac{3|I| \log Z}{\pi^2} + o(1)\right) Q^2\right). \quad (15)$$

Thus no such rational conjugacy can satisfy a denominator estimate $\text{den } G(p/q) \ll q^A$ for any fixed A .

Proof. For a fixed $I \Subset J$, the map $t \mapsto (\phi(t), G(t))$ injects $\mathbb{Q} \cap I$ into the orbit. The formula for the continuous extension follows pointwise from $S_k = R_k^{\log_2 3}$. Apply Theorem 4.2 to the second coordinate. $\square$

Corollary 4.5 (Bounded multiplicity version). *If every orbit point has at most M preimages under a map $\Phi : \mathbb{Q} \cap I \rightarrow \{(R_k, S_k)\}$, where M is fixed, then the constants on the right of (13) and (14) may be divided by M . In particular, polynomial denominator growth is still impossible.*

Proof. The N_Q input rationals occupy at least N_Q/M distinct orbit indices. Repeat the proof of Theorem 4.2. $\square$

Proposition 4.6 (Prime-level denominator explosion). *Let $\Phi : \mathbb{Q} \cap I \rightarrow \{(R_k, S_k) : k \geq 0\}$ be injective. For every sufficiently large prime q , put*

$$E_2(q) = \max_{\substack{p/q \in I \\ 1 \leq p < q}} \text{den}(\pi_1 \Phi(p/q)), \quad E_3(q) = \max_{\substack{p/q \in I \\ 1 \leq p < q}} \text{den}(\pi_2 \Phi(p/q)).$$

Then

$$\log E_2(q) \geq (|I| \log W + o(1))q, \quad (16)$$

$$\log E_3(q) \geq (|I| \log Z + o(1))q. \quad (17)$$

Thus even when one fixes a single prime denominator q , at least one output denominator is exponential in q .

Proof. For prime q , every $1 \leq p < q$ is coprime to q , and the number of p with $p/q \in I$ is $|I|q + O(1)$. Their images have distinct orbit indices, so one index is at least $|I|q + O(1)$. Apply the denominator lower bounds in Theorem 3.1. $\square$

Corollary 4.7 (Enumeration form of the entropy mismatch). *Enumerate the rationals in I in nondecreasing height and inject them into the orbit. Among the first N inputs, some image has each coordinate denominator at least*

$$\frac{1}{2}W^{N-1} \quad \text{and} \quad \frac{1}{3}Z^{N-1},$$

respectively. Since the N th rational in a height ordering has height $\asymp_I \sqrt{N}$, the optimal possible denominator distortion is already $\exp(cH^2)$.

Proof. Distinct images use N distinct nonnegative indices, one of which is at least $N - 1$. The height assertion follows by inverting the Farey estimate (12). $\square$

4.1 Interpretation

There are $\asymp Q^2$ rationals of height at most Q in any fixed interval, but only $\asymp \log H$ points of the conditional orbit of height at most H . An injective map from the first set to the second must therefore solve

$$Q^2 \lesssim \log H,$$

forcing $H \gtrsim e^{cQ^2}$. This is an arithmetic entropy mismatch: the Farey set has quadratic height entropy, while the cyclic conditional orbit has logarithmic height entropy.

The result is not a contradiction to the existence of the orbit. It proves instead that any attempt to “fill in” the orbit by a rational parameter must pay a denominator cost far outside the range where known smooth-rigidity theorems operate.

5 The 2026 rational-map rigidity theorem and its exact boundary

5.1 The external theorem

The following is the form of the recent theorem needed for comparison.

Theorem 5.1 (Lelis–Moreira–Silva, 2026; simplified statement). *Let $k \geq 1$, let Ω be a connected neighborhood of 0, and let $f \in C^{2k+1}(\Omega)$. Assume*

- (i) $f^{(j)}(0) \in \mathbb{Q}$ for $0 \leq j \leq 2k$;
- (ii) $f(\Omega \cap \mathbb{Q}) \subseteq \mathbb{Q}$;
- (iii) $\text{den } f(p/q) \ll q^k$ for rationals $p/q \in \Omega$ with sufficiently large q .

Then f is a rational function with rational coefficients of degree at most k .

The proof combines a rational Padé-type construction from the rational jet with a subunit-locking argument: after subtraction of the rational approximant, the residual is $O(q^{-(2k+1)})$ while its denominator is $O(q^{2k})$, so a nonzero rational residual is impossible for large q [5].

5.2 Why the conditional power curve misses both decisive hypotheses

Put

$$\vartheta = \log_2 3 = \frac{\log 3}{\log 2}, \quad f(t) = t^\vartheta.$$

The number ϑ is transcendental: it is irrational, and if it were algebraic then Gelfond–Schneider would make $2^\vartheta = 3$ transcendental.

Under a counterexample, $f(R_k) = S_k \in \mathbb{Q}$ on a dense set of rational arguments. This looks close to Theorem 5.1, but it is not close in height geometry.

First, the theorem requires every rational argument in a neighborhood, not a dense cyclic subset. The orbit contributes only $O(\log H)$ rational graph points of height at most H . Second, any reparametrization that maps all rationals injectively into the orbit has $\exp(cq^2)$ denominator distortion by Theorem 4.2, rather than the required q^k .

Third, the rational-jet condition fails already in order one.

Proposition 5.2 (Transcendental tangent at every rational graph point). *Let $a, b \in \mathbb{Q}_{>0}$ satisfy $b = a^\vartheta$. Then*

$$f'(a) = \vartheta \frac{b}{a}$$

is transcendental. Consequently, if $R \in \mathbb{Q}(t)$ is regular at a and $R(a) = b$, then $f - R$ has a simple zero at a ; in particular, it cannot have contact order at least two.

Proof. The factor b/a is a nonzero rational and ϑ is transcendental, so $\vartheta b/a$ is transcendental. On the other hand, $R'(a) \in \mathbb{Q}$. Therefore $f'(a) - R'(a) \neq 0$. $\square$

Corollary 5.3 (No direct Lelis–Moreira–Silva reduction). *Neither of the following can turn a hypothetical Alaoglu–Erdős orbit into an application of Theorem 5.1:*

- (a) *translating the power function around one of its rational graph points;*
- (b) *injectively reparametrizing the entire rational set of a neighborhood by orbit points with polynomial denominator growth.*

Proof. In (a), the translated function has a transcendental first derivative at the origin by Proposition 5.2. In (b), polynomial denominator growth contradicts Theorem 4.2. $\square$

Remark 5.4 (Density is not the missing theorem). A continuous function agreeing with t^ϑ on the dense orbit agrees with it on the whole interval. Thus a continuous extension that also mapped every rational to a rational would force $t^\vartheta \in \mathbb{Q}$ for every rational t in an interval. That is already incompatible with the Six Exponentials Theorem after choosing three multiplicatively independent rational bases. The hard issue is not uniqueness of the continuous extension; it is producing rational values away from the rank-two orbit.

6 Orbit-adapted subunit locking and the exponential denominator tax

The Lelis–Moreira–Silva proof wins because a residual of order $q^{-(2k+1)}$ is a rational with denominator only $O(q^{2k})$. On the conditional orbit, the geometric recurrence scale is polynomial in the orbit index, but the rational denominators are exponential in that index. We now make this comparison exact.

6.1 Return times near the rational point $(1, 1)$

Let p_j/q_j be the continued-fraction convergents of α . Infinitely many are below α , so for an infinite subsequence

$$0 < \delta_j := q_j \alpha - p_j < \frac{1}{q_{j+1}} < \frac{1}{q_j}. \quad (18)$$

Because W and 2 are multiplicatively independent, a standard two-logarithm consequence of Baker’s theorem [2] gives constants $c_W > 0$ and $\kappa_W > 1$ such that

$$|q\alpha - p| \geq c_W q^{-\kappa_W} \quad (q \geq 2, p \in \mathbb{Z}). \quad (19)$$

Indeed, if $|q\alpha - p| \geq 1$ the claim is trivial after shrinking c_W ; otherwise $p = O_W(q)$. After multiplying by $\log 2$, the left side is then the nonzero linear form $q \log W - (rq + p) \log 2$, whose integer coefficients

have size $O_W(q)$. In particular, the return scale in (18) is polynomial, never exponential, in the orbit index; also $q_{j+1} \ll_W q_j^{\kappa_W}$.

For this subsequence,

$$0 < R_{q_j} - 1 = 2^{\delta_j} - 1 \ll \frac{1}{q_j}, \quad (20)$$

$$0 < S_{q_j} - 1 = 3^{\delta_j} - 1 \ll \frac{1}{q_j}. \quad (21)$$

The implied constants in these two upper bounds are absolute.

Lemma 6.1 (Height of a fixed rational expression). *Let $F \in \mathbb{Q}(X, Y)$ be regular on a neighborhood of the compact arc $\mathcal{A}$. There is a constant $C_F > 0$ such that, whenever $F(R_k, S_k) \neq 0$,*

$$|F(R_k, S_k)| \geq e^{-C_F k}. \quad (22)$$

Proof. Write $F = A/B$ with $A, B \in \mathbb{Z}[X, Y]$ after clearing a fixed common denominator. Let d be a bound for the total degrees and H_F a bound for the coefficients. Standard height inequalities give

$$H(F(R_k, S_k)) \leq C'_F H(R_k)^d H(S_k)^d \leq C'_F (WZ)^{dk}.$$

A nonzero rational number of height at most H has absolute value at least H^{-1} . Absorbing the fixed factor C'_F into the exponent proves (22). $\square$

Theorem 6.2 (Fixed algebraic contact has only polynomial strength). *Let $0 \neq P \in \mathbb{Q}[X, Y]$ satisfy $P(1, 1) = 0$, and define the finite positive integer*

$$\sigma(P) = \text{ord}_{t=0} P(2^t, 3^t).$$

Along the subsequence (18), the values $P(R_{q_j}, S_{q_j})$ are nonzero for all sufficiently large j , and there are constants $c_P, C_P > 0$ such that

$$c_P q_j^{-\kappa_W \sigma(P)} \leq |P(R_{q_j}, S_{q_j})| \leq C_P q_j^{-\sigma(P)}. \quad (23)$$

They also satisfy the weaker height-theoretic estimate

$$|P(R_{q_j}, S_{q_j})| \geq e^{-C'_P q_j}. \quad (24)$$

If P has algebraic multiplicity s at $(1, 1)$, then $\sigma(P) \geq s$. Consequently, every fixed polynomial contact certificate has only polynomial smallness in the orbit index and cannot activate an exponential-denominator subunit argument.

Proof. Put $F(t) = P(2^t, 3^t)$. This analytic function is not identically zero. Indeed, writing $P(X, Y) = \sum c_{ab} X^a Y^b$ gives

$$F(t) = \sum c_{ab} e^{(a \log 2 + b \log 3)t};$$

the exponents are distinct because $\log 2$ and $\log 3$ are linearly independent over $\mathbb{Q}$, and distinct exponential functions are linearly independent. Hence $\sigma(P) < \infty$, and for sufficiently small positive

t ,

$$c'_P t^{\sigma(P)} \leq |F(t)| \leq C'_P t^{\sigma(P)}.$$

Now $P(R_{q_j}, S_{q_j}) = F(\delta_j)$. Combining (18) and (19) proves (23), and also shows eventual nonvanishing. The estimate (24) follows independently from Lemma 6.1. Finally, a Taylor expansion of P in $(X - 1, Y - 1)$ shows that algebraic multiplicity s forces $F(t) = O(t^s)$, whence $\sigma(P) \geq s$. $\square$

The important point is now two-sided: Baker’s theorem prevents exponentially close returns, and analyticity converts every fixed contact order into a fixed power of that return. Thus the actual certificate is polynomially, not exponentially, small. The elementary rational lower bound (24) is far too weak to collide with it.

6.2 A general jet-budget inequality

The scale comparison can be abstracted.

Proposition 6.3 (Orbit jet budget). *Suppose a family of nonzero rational certificates Ξ_N satisfies*

$$\text{den}(\Xi_N) \leq \exp(A_N N) \tag{25}$$

and an analytic construction gives

$$|\Xi_N| \leq C_N N^{-d_N}. \tag{26}$$

A contradiction from the elementary lower bound $|\Xi_N| \geq \text{den}(\Xi_N)^{-1}$ requires

$$d_N \log N > A_N N + \log C_N. \tag{27}$$

In particular, fixed $A_N = A > 0$, fixed $d_N = d$, and subexponential C_N can never suffice. If A_N stays bounded away from zero and $C_N = e^{o(N)}$, then necessarily

$$d_N \geq (A_N + o(1)) \frac{N}{\log N}. \tag{28}$$

Proof. A contradiction requires the analytic upper bound to be smaller than the arithmetic lower bound:

$$C_N N^{-d_N} < e^{-A_N N}.$$

Taking logarithms gives (27); the remaining statements are immediate. $\square$

Audit warning 6.4. For actual variable-degree determinants, A_N usually grows with the degree because every additional node carries an exponential height cost. Thus (28) is only a necessary benchmark, not a promise that order $N/\log N$ is enough. The supplied report’s large determinant audits analyze precisely this extra coefficient and denominator growth.

6.3 Why the 2026 one-power win disappears

On the rational grid n/q , a zero of order $2k + 1$ contributes $q^{-(2k+1)}$, while the cleared denominator in Theorem 5.1 is only q^{2k} . The exponent gap of one locks the residual to zero.

On the conditional orbit, a return of index N is only polynomially close to a previous point, whereas the denominator is $\asymp W^N$ or $\asymp Z^N$. Any fixed-order Taylor gain is therefore polynomial, while the rational lower bound is exponentially small. The inequality points in the wrong direction by an exponential margin. This is the exact reason the new rational-map theorem does not cross the Alaoglu–Erdős wall.

7 Further proof attempts pushed to explicit failure points

This section records several routes explored after the supplied report. The purpose is not to accumulate reformulations, but to identify the first unjustified inference in each route.

7.1 Route A: use a smooth rational parametrization

Because the orbit is countable and dense, it has the same order type as $\mathbb{Q} \cap I$; an order isomorphism extends to an increasing homeomorphism. One might hope to choose the homeomorphism smoothly and then invoke rational-map rigidity.

Exact failure. Smoothness is irrelevant to the counting obstruction. Corollary 4.4 forces denominator growth $\exp(cq^2)$ for every injective rational parametrization. This is vastly larger than the polynomial bounds in all known finite-smoothness rigidity theorems.

7.2 Route B: extend rationality from a dense set

The identity theorem says that two analytic functions agreeing on a dense set agree everywhere. It does not say that an analytic function taking rational values on a dense set is a rational function. Transcendental entire functions mapping $\mathbb{Q}$ into $\mathbb{Q}$ have been known since the nineteenth century, and much stronger modern constructions exist.

Exact failure. Rationality is not a closed condition in the real topology. A limit of rational values can be arbitrary. The missing input must control heights or denominators, not merely density.

7.3 Route C: rational Padé approximation at an orbit point

At a rational graph point (a, b) , one can choose $R \in \mathbb{Q}(t)$ with $R(a) = b$ and hope for high contact with t^ϑ .

Exact failure. Proposition 5.2 gives contact order exactly one. A higher Padé order requires coefficients involving ϑ ; then evaluation at a rational orbit point is no longer rational, and the arithmetic subunit argument disappears. Bivariate rational polynomials can attain higher algebraic multiplicity, but Theorem 6.2 shows that every fixed order remains exponentially underpowered.

7.4 Route D: force p -adic continuity

The hypotheses define a group homomorphism on the rank-two subgroup

$$\langle 2, 3 \rangle \subset \mathbb{Q}_{>0}, \quad 2^a 3^b \mapsto m^a n^b.$$

If this homomorphism extended continuously to a suitable p -adic closure, p -adic logarithms would impose an additional linear relation and might create a mixed archimedean–nonarchimedean determinant.

Exact failure. A homomorphism from a dense countable subgroup of a compact p -adic group need not be continuous. Integrality on the positive semigroup only gives p -adic boundedness there; it gives no control on sequences with signed exponents $2^{a_j} 3^{b_j} \rightarrow 1$. The required assertion is therefore a new continuity theorem, not a formal consequence of integrality.

A proof-producing version of this route must establish the implication

$$2^{a_j} 3^{b_j} \rightarrow 1 \text{ in } \mathbb{Q}_p \implies m^{a_j} n^{b_j} \rightarrow 1 \text{ in } \mathbb{Q}_p \quad (29)$$

for at least one finite place and for all integer sequences (a_j, b_j) . No argument found in the supplied report, in the inspected literature, or in the present work proves (29).

7.5 Route E: equidistribution and shrinking targets

The rotation $k\alpha \bmod 1$ is equidistributed. Exact integrality, however, asks for simultaneous hits on grids whose mesh is exponential in the block height. Equidistribution controls fixed intervals and, with Diophantine estimates, some polynomially shrinking intervals. It does not force a hit in an exponentially shrinking target. In fact, Theorem 3.1 shows that a counterexample would itself generate exactly the required exceptional orbit, so generic metric heuristics cannot rule it out.

7.6 Route F: treat the Matrix Coefficient Conjecture as proved

For the matrix

$$M = \begin{pmatrix} \log 2 & \log 3 \\ \log m & \log n \end{pmatrix},$$

the Matrix Coefficient Conjecture would turn singularity into a rational row–column coefficient relation. In size two this is equivalent to Four Exponentials [4].

Exact failure. The relevant matrix-coefficient statement is a conjecture, not a proved theorem. Dasgupta and Kakde prove important structural implications and an analogue of a Masser theorem, but explicitly leave the implication needed to complete their strategy open. Citing the paper as an unconditional determinant-nonvanishing theorem would be a misreading.

8 Conditional amplifiers that really would prove the conjecture

The following criteria separate useful targets from equivalent restatements.

Criterion 8.1 (A third algebraic direction). *Suppose the hypotheses $2^x, 3^x \in \mathbb{Z}$ imply the existence of a positive algebraic number c such that*

- (i) c^x is algebraic;
- (ii) $\log c$ is linearly independent over $\mathbb{Q}$ from $\log 2, \log 3$.

Then $x \in \mathbb{Z}$.

Proof. If $x \notin \mathbb{Q}$, then $1, x$ are $\mathbb{Q}$ -independent. The three logarithms are $\mathbb{Q}$ -independent by assumption. The Six Exponentials Theorem says that among the six numbers

$$2, 3, c, 2^x, 3^x, c^x$$

at least one is transcendental, contrary to algebraicity. Hence $x \in \mathbb{Q}$, and Proposition 2.1 gives $x \in \mathbb{Z}$. $\square$

The difficulty is not proving the criterion but manufacturing c . Every base generated multiplicatively from $2, 3, m, n$ remains in the same rank-two logarithmic span.

Criterion 8.2 (Denominator-tame thickening). *Assume a counterexample produces, for some interval $J \subset (1, 2)$, a set $\mathcal{T} \subset \mathbb{Q} \cap J$ such that*

- (i) $t^\vartheta \in \mathbb{Q}$ for every $t \in \mathcal{T}$;
- (ii) $\#\{t \in \mathcal{T} : H(t), H(t^\vartheta) \leq H\} \gg H^\delta$ for some $\delta > 0$;
- (iii) the denominators obey a uniform polynomial transfer law.

Then determinant or rational-point rigidity methods enter a genuinely stronger regime than the conditional orbit.

This is deliberately stated as an amplifier rather than a theorem. The conditional orbit has only $\asymp \log H$ points, and Theorem 4.2 shows that a mere reparametrization cannot create the required height density. A successful proof must create new rational graph points by an arithmetic operation, not relabel the old ones.

Research problem 8.3 (Prime-supported sparse rigidity). Let $0 < \alpha < 1$ be irrational. Rule out the simultaneous existence of integers $r \geq 1$, $s \geq 0$, $W > 1$, and $Z > 1$ satisfying

$$W \text{ odd}, \quad 3 \nmid Z, \quad r + \alpha = \log_2 W, \quad s + \alpha = \log_3 Z.$$

Equivalently, rule out a dense cyclic sequence on $\{(2^t, 3^t) : 0 \leq t < 1\}$ whose reduced coordinate denominators are exactly

$$2^{\lfloor k \log_2 W \rfloor} \quad \text{and} \quad 3^{\lfloor k \log_3 Z \rfloor},$$

with their floor fluctuations synchronized by the same irrational rotation.

Problem 8.3 deliberately isolates the extra arithmetic structure that ordinary rational-point counting ignores. A useful solution would have to exploit the synchronized pure-prime denominator laws, rather than merely recast the original logarithmic equality.

Research problem 8.4 (One finite-place continuity theorem). Under the counterexample hypotheses, prove (29) for one prime p at which the logarithms involved are nondegenerate, and combine the resulting p -adic linear dependence with the real determinant (1).

This route is attractive because it would add a genuinely independent local direction. It is also dangerous: continuity is precisely the unproved content, so it must not be hidden inside notation such as $x \log_p 2$ for a real transcendental x .

Research problem 8.5 (Orbit-scale rational rigidity). Find a rational-map rigidity theorem whose hypotheses use the exact cyclic denominator law (6)–(7) rather than requiring all rational arguments, and whose conclusion excludes the transcendental power function. The theorem must remain effective when the rational-point count is only $c \log H$.

Theorem 4.2 shows why a theorem proved by first replacing the orbit with all rationals cannot work with polynomial heights. Any successful result must be intrinsically sparse and prime-sensitive.

9 A sharpened logical map of the remaining wall

The following table records what each major route would need beyond currently established results.

Route	Established input	Missing proof-producing step
Four Exponentials	Exact 2×2 logarithmic determinant	Four Exponentials itself, or a special integral case
Matrix coefficients	Size-two equivalence and partial strategy	Open matrix-coefficient implication
Smooth rational maps	2026 C^{2k+1} rigidity at denominator q^k	All-rational coverage and rational jet; orbit reparametrization costs e^{cq^2}
Polynomial / Padé	Exact rational contact and denominator clearing	Variable-complexity gain overcoming exponential height tax
Rational-point counting	Exact count $1 + \lfloor \log H / \log C_0 \rfloor$	Prime-supported theorem effective at logarithmic count
p -adic logarithms	Local logarithmic rank tools	Continuity of the real-power homomorphism in one finite place
Third-base method	Six Exponentials Theorem	Construct one algebraic exponential outside the rank-two span

The common pattern is clear. The hypotheses provide two algebraic directions and one real transcendental scaling. Every successful existing theorem wants either a third algebraic direction, a rational jet, a polynomially dense rational set, or an independent local continuity statement. The conditional orbit supplies none of these automatically.

10 Conclusion

The work above does not prove the Alaoglu–Erdős conjecture. It does produce a new and rigorous continuation of the supplied report:

1. The compact counterexample orbit has exact heights

$$H(R_k) = W^k, \quad H(S_k) = Z^k,$$

not merely exponential upper and lower bounds.

2. Every injection from all rationals in an interval into that orbit has denominator distortion

$$\exp\left(\left(\frac{3|I|\log W}{\pi^2} + o(1)\right)Q^2\right) \quad \text{and} \quad \exp\left(\left(\frac{3|I|\log Z}{\pi^2} + o(1)\right)Q^2\right).$$

This rules out every polynomially tame reparametrization.

3. The July 2026 smooth rational-map rigidity theorem cannot be applied directly: the orbit is height-sparse, every tame thickening is impossible by the entropy theorem, and the power curve has a transcendental first derivative at every rational graph point.
4. Fixed algebraic contact gives only polynomial smallness in a recurrence index, while rational denominator clearing gives an exponential lower bound. These estimates are compatible, so a fixed-order subunit argument cannot close the problem.
5. A genuine proof still requires an amplifier: a third algebraic base, a prime-sensitive rigidity theorem at logarithmic point density, an independently proved p -adic continuity statement, or a new determinant theorem strong enough to establish the needed size-two matrix coefficient relation.

The central new lesson is quantitative. The gap between the known 2026 rational-map rigidity regime and the Alaoglu–Erdős orbit is not a missing constant or one more finite-difference estimate. It is an entropy gap between polynomially many rational inputs and logarithmically many rational graph points, forcing e^{cQ^2} denominator distortion. Any future proof using rational approximation must exploit the synchronized pure-prime denominators directly rather than trying to replace the orbit by an ordinary rational grid.

A Farey counting on a fixed interval

For completeness, we recall the standard estimate used in Theorem 4.2. For a fixed interval $I = [a, b] \subset (0, 1)$ and a fixed q , the number of p such that $p/q \in I$ and $\gcd(p, q) = 1$ is

$$|I|\varphi(q) + O(\tau(q)),$$

where φ is Euler’s totient and τ the divisor function. This follows by Möbius inversion:

$$\sum_{\substack{aq \leq p \leq bq \\ (p,q)=1}} 1 = \sum_{d|q} \mu(d) \left(\frac{|I|q}{d} + O(1) \right) = |I|\varphi(q) + O(\tau(q)).$$

Summing over $q \leq Q$ and using

$$\sum_{q \leq Q} \varphi(q) = \frac{3}{\pi^2} Q^2 + O(Q \log Q), \quad \sum_{q \leq Q} \tau(q) = O(Q \log Q),$$

gives

$$\#\mathcal{F}_I(Q) = \frac{3|I|}{\pi^2} Q^2 + O_I(Q \log Q).$$

Only the quadratic main term is needed for the entropy theorem.

B A direct denominator-count formulation

Let

$$\Delta_k = \max\{\text{den}(R_k), \text{den}(S_k)\}, \quad C_0 = \max\{W, Z\}.$$

Theorem 3.1 gives

$$\frac{C_0^k}{3} < \Delta_k < C_0^k.$$

Therefore the denominator counting function satisfies

$$1 + \left\lfloor \frac{\log D}{\log C_0} \right\rfloor \leq \#\{k \geq 0 : \Delta_k \leq D\} \leq 1 + \left\lfloor \frac{\log(3D)}{\log C_0} \right\rfloor. \quad (30)$$

Thus the orbit has an exact logarithmic denominator entropy with leading constant $1/\log C_0$. The Farey set has quadratic height entropy with leading constant $3|I|/\pi^2$. Equation (15) is precisely the inequality obtained by comparing these two entropies.

C Literature note as of August 21, 2026

The literature check for this continuation focused on developments capable of changing the logical status of the problem.

- The classical three-base result follows from Six Exponentials; replacing three bases by two remains the difficult case [8].
- Dasgupta–Kakde’s Matrix Coefficient Conjecture is explicitly conjectural, and its 2×2 case is equivalent to Four Exponentials [4].
- Massold’s 2025 work develops new transcendence-degree estimates but does not prove the Four Exponentials statement needed here [6].

- Lelis–Moreira–Silva’s July 2026 theorem supplies the newest directly relevant rational-map rigidity result found in the search. Its all-rational, rational-jet, and polynomial-denominator hypotheses are substantially stronger than the data supplied by the compact conditional orbit [5].

No inspected paper supplied an unconditional proof of the two-base conjecture. This negative literature statement is not used as a mathematical premise; it explains why the report distinguishes proved auxiliary theorems from the still-open final implication.

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